

Tree Stethoscope Engineers

Week 4 | October 10, 2026

Ranger Name: _____

What We Learned in Class

Emerald ash borer larvae chew tunnels just under the bark and cut the tree's **straws** (xylem and phloem). By the time we can **see** clues like a thin top or D-shaped holes, the tree is already very sick. Today people use **bug traps** to find where the beetle is and **injections** to protect healthy trees. Our idea, the **Tree Stethoscope**, tries to **hear** the larvae early.

Part 1 Pick Materials for Our Tree Stethoscope

Our device will live **outside on a tree** in rain, sun, and snow, and it **must not hurt the tree**. Choose a material for each part and say why.

Ideas: plastic (3D printed), metal, rubber, cloth, wood, LEGO

Part	Material	Why?
Case keeps the M5Stack dry		
Strap holds it on the tree		
Sensor tip touches the tree to listen		

Part 2 Read a Real Research Paper

College students at Loyola University Chicago built a sensor to **listen** to EAB larvae inside young ash trees. With a parent, look at the title, pictures, and headings.

"Acoustic Monitoring of Emerald Ash Borer Larvae in Ash Tree Saplings" (2023)

What problem were the students trying to solve?

What did they use to listen to the larvae?

One way their sensor is like our Tree Stethoscope, and one way it is different:

Think Like an Engineer

Part 3–4 | Power and where it works

Part 3 Power

Our device has to work on a tree for a long time. Where could it get power?

☐ Battery ☐ Solar panel ☐ Plug into the wall ☐ Other: _____

How could we make the battery last longer? Hint: does it need to listen all day and all night?

Part 4 Where Would It Work?

Where would our Tree Stethoscope be most useful? Why?

☐ City park ☐ Street tree ☐ Backyard ☐ Forest

What could make it hard to hear the larvae? Hint: wind, cars, birds, people?

Who would want to use it?
